

Matrix functions and stability

Polynomial conditioning of the cubic C1 spline Schrödinger Toeplitz family

Difficulty: hard **Importance:** interesting to specialist**Status:** Open**Topic:** Block Toeplitz systems and discretization stability

Rating rationale: The fixed explicit band structure makes this a hard asymptotic question, but available symbol criteria leave the relevant regime undecided; its immediate importance is to specialists in structured discretization stability.

Statement

Define real 2×2 matrices by

$$\begin{aligned} B_{-1} &= \frac{3}{40} \begin{pmatrix} -1 & 0 \\ -5 & 0 \end{pmatrix}, & B_0 &= \frac{3}{40} \begin{pmatrix} 0 & -5 \\ 16 & -5 \end{pmatrix}, \\ B_1 &= \frac{3}{40} \begin{pmatrix} -5 & 16 \\ -5 & 0 \end{pmatrix}, & B_2 &= \frac{3}{40} \begin{pmatrix} 0 & -5 \\ 0 & -1 \end{pmatrix}, \\ C_{-1} &= \frac{1}{80} \begin{pmatrix} 1 & 0 \\ 7 & 0 \end{pmatrix}, & C_0 &= \frac{1}{80} \begin{pmatrix} 24 & 7 \\ 0 & 25 \end{pmatrix}, \\ C_1 &= \frac{1}{80} \begin{pmatrix} -25 & 0 \\ -7 & -24 \end{pmatrix}, & C_2 &= \frac{1}{80} \begin{pmatrix} 0 & -7 \\ 0 & -1 \end{pmatrix}. \end{aligned}$$

Set $B_k = C_k = 0$ for all other integer indices. For a real parameter $\rho > 0$ and $n \geq 1$, form the $2n \times 2n$ complex block Toeplitz matrix

$$H_n(\rho) = (iB_{j-k} - \rho C_{j-k})_{j,k=0}^{n-1}, \quad i^2 = -1.$$

Use the spectral condition number $\kappa_2(H) = \|H\|_2 \|H^{-1}\|_2$ for nonsingular H and $\kappa_2(H) = +\infty$ for singular H .

Question (explicit asymptotic formulation of the source's polynomial-growth question). For every fixed $\rho > 0$, do there exist constants $K_\rho > 0$, $\alpha_\rho \geq 0$, and $n_\rho \in \mathbb{N}$, independent of n , such that

$$\kappa_2(H_n(\rho)) \leq K_\rho n^{\alpha_\rho} \quad \text{for every } n \geq n_\rho?$$

The parameter remains fixed as $n \rightarrow \infty$; no uniform-in- ρ bound or particular exponent is asserted. Eventual invertibility is part of the question. This is the exact pure Toeplitz family denoted $\tilde{\mathbf{S}}_n^{(3,1)}(\rho)$ in the source; n here counts its 2×2 blocks. It has no additional corner corrections.

Numerical significance

These matrices arise from cubic splines with C^1 continuity in time for a space-time Schrödinger discretization. The source proves a determinant-root classification of the associated polynomial symbol but expressly leaves polynomial growth of these condition numbers open. A bound would explain the numerical stability seen for this intermediate spline regularity.

References and status check

- M. Bogoya, A. Böttcher, M. Ferrari, S. M. Grudsky and S. Serra-Capizzano, *Condition numbers of block Toeplitz matrices and stability of space-time IgA approximations for the wave and Schrödinger equations*, [arXiv:2608.24151v1](#), 25 August 2026. **Full text:** equation (5.22) gives B_k ; §5.3, equation (5.32) and the following coefficient display give the family and C_k ; the paragraph after Figure 9 and Remark 4.3 state the unresolved polynomial-growth issue.
- M. Ferrari and S. Gómez, *A matrix-based approach to the stability of a space-time isogeometric method for the linear Schrödinger equation*, [arXiv:2506.18859v2](#) (5 May 2026), abstract and introduction. Its analyzed method uses splines of maximal regularity, a different case from cubic C^1 splines.

On 2026-09-10, checked the complete August 2026 source, its version history, and targeted identifier/title/correction searches. Only v1 was listed, and no later proof or counterexample was found. This short, bounded follow-up search does not establish exhaustive openness. The source's more general claim involving only determinant-root counts is not needed for this particular, explicitly stated family.